EVERY 1-MORPHISM OF FIBRED CATEGORIES FACTORS THROUGH A FULLY FAITHFUL ONE WITH A LEFT ADJOINT

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Editorial note. Counted span is the author's own proof environment, source lines 6900-6976, following the statement at 6881-6898.

Editorial note. The adjointness bijection is marked 'follows immediately from the definitions'; the morphism description it unfolds is given in-span at 6906-6908.

Editorial note. The proof uses no lemma references; only the chapter definitions.

Allowed standard prerequisites (editorial).

- Fibred categories and strongly cartesian morphisms. Definitions of fibred category and strongly cartesian morphism in this chapter. The verification that X' is fibred and that u, v, w are 1-morphisms of fibred categories runs on these definitions; F being a 1-morphism of fibred categories makes F(h*x) -> F(x) strongly cartesian.
- Choice of pullbacks for the structure functors. Made at proof lines 6911-6912 following the chapter convention. Fixes the strongly cartesian lifts h*x -> x and h*y -> y used for the comparison morphism c and the lift (a, c).

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of fibred categories over $\mathcal{C}$. There exist 1-morphisms of fibred categories over $\mathcal{C}$*

$$\mathcal{X} \underset{w}{\overset{u}{\rightleftarrows}} \mathcal{X}' \xrightarrow{v} \mathcal{Y}$$

such that $F = v \circ u$ and such that

(1) *$u : \mathcal{X} \to \mathcal{X}'$ is fully faithful,*
(2) *w is left adjoint to u, and*
(3) *$v : \mathcal{X}' \to \mathcal{Y}$ is a fibred category.*

Proof. Denote $p : \mathcal{X} \to \mathcal{C}$ and $q : \mathcal{Y} \to \mathcal{C}$ the structure functors. We construct $\mathcal{X}'$ explicitly as follows. An object of $\mathcal{X}'$ is a quadruple (U, x, y, f) where $x \in \mathrm{Ob}(\mathcal{X}_U)$, $y \in \mathrm{Ob}(\mathcal{Y}_U)$ and $f : y \to F(x)$ is a morphism in $\mathcal{Y}_U$. A morphism $(a, b) : (U, x, y, f) \to (U', x', y', f')$ is given by $a : x \to x'$ and $b : y \to y'$ with $p(a) = q(b) : U \to U'$ and such that $f' \circ b = F(a) \circ f$.

Let us make a choice of pullbacks for both p and q and let us use the same notation to indicate them. Let (U, x, y, f) be an object and let $h : V \to U$ be a morphism. Consider the morphism $c : (V, h^*x, h^*y, h^*f) \to (U, x, y, f)$ coming from the given strongly cartesian maps $h^*x \to x$ and $h^*y \to y$. We claim c is strongly cartesian in $\mathcal{X}'$ over $\mathcal{C}$. Namely, suppose we are given an object (W, x', y', f') of $\mathcal{X}'$, a morphism $(a, b) : (W, x', y', f') \to (U, x, y, f)$ lying over $W \to U$, and a factorization $W \to$

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$V \to U$ of $W \to U$ through h. As $h^*x \to x$ and $h^*y \to y$ are strongly cartesian we obtain morphisms $a' : x' \to h^*x$ and $b' : y' \to h^*y$ lying over the given morphism $W \to V$. Consider the diagram

$$\begin{array}{ccccc} y' & \longrightarrow & h^*y & \longrightarrow & y \\ {\scriptstyle f'}\downarrow & & {\scriptstyle h^*f}\downarrow & & {\scriptstyle f}\downarrow \\ F(x') & \longrightarrow & F(h^*x) & \longrightarrow & F(x) \end{array}$$

The outer rectangle and the right square commute. Since F is a 1-morphism of fibred categories the morphism $F(h^*x) \to F(x)$ is strongly cartesian. Hence the left square commutes by the universal property of strongly cartesian morphisms. This proves that $\mathcal{X}'$ is fibred over $\mathcal{C}$.

The functor $u : \mathcal{X} \to \mathcal{X}'$ is given by $x \mapsto (p(x), x, F(x), \mathrm{id})$. This is fully faithful. The functor $\mathcal{X}' \to \mathcal{Y}$ is given by $(U, x, y, f) \mapsto y$. The functor $w : \mathcal{X}' \to \mathcal{X}$ is given by $(U, x, y, f) \mapsto x$. Each of these functors is a 1-morphism of fibred categories over $\mathcal{C}$ by our description of strongly cartesian morphisms of $\mathcal{X}'$ over $\mathcal{C}$. Adjointness of w and u means that

$$\mathrm{Mor}_{\mathcal{X}}(x, x') = \mathrm{Mor}_{\mathcal{X}'}((U, x, y, f), (p(x'), x', F(x'), \mathrm{id})),$$

which follows immediately from the definitions.

Finally, we have to show that $\mathcal{X}' \to \mathcal{Y}$ is a fibred category. Let $c : y' \to y$ be a morphism in $\mathcal{Y}$ and let (U, x, y, f) be an object of $\mathcal{X}'$ lying over y. Set $V = q(y')$ and let $h = q(c) : V \to U$. Let $a : h^*x \to x$ and $b : h^*y \to y$ be the strongly cartesian morphisms covering h. Since F is a 1-morphism of fibred categories we may identify $h^*F(x) = F(h^*x)$ with strongly cartesian morphism $F(a) : F(h^*x) \to F(x)$. By the universal property of $b : h^*y \to y$ there is a morphism $c' : y' \to h^*y$ in $\mathcal{Y}_V$ such that $c = b \circ c'$. We claim that

$$(a, c) : (V, h^*x, y', h^*f \circ c') \longrightarrow (U, x, y, f)$$

is strongly cartesian in $\mathcal{X}'$ over $\mathcal{Y}$. To see this let (W, x_1, y_1, f_1) be an object of $\mathcal{X}'$, let $(a_1, b_1) : (W, x_1, y_1, f_1) \to (U, x, y, f)$ be a morphism and let $b_1 = c \circ b_1'$ for some morphism $b_1' : y_1 \to y'$. Then

$$(a_1', b_1') : (W, x_1, y_1, f_1) \longrightarrow (V, h^*x, y', h^*f \circ c')$$

(where $a_1' : x_1 \to h^*x$ is the unique morphism lying over the given morphism $q(b_1') : W \to V$ such that $a_1 = a \circ a_1'$) is the desired morphism. $\square$